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Course code: CSE4004

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Slot: L37+L38

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Lab Sheet 6: MongoDB Basic commands

1. Use MongoDB to implement the following DB operations

1. Create a database called 'vehicles' and write a MongoDB query to select database as "vehicles".

```
test> use vehicles
switched to db vehicles
```

2. Write a MongoDB query to display all the databases.

```
vehicles> show dbs
admin       40.00 KiB
config      96.00 KiB
local       64.00 KiB
vehicles   144.00 KiB
```

3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.

```
vehicles> db.createCollection("two_wheelers",{capped:true,size:50000})
{ ok: 1 }
vehicles> db.createCollection("four_wheelers")
{ ok: 1 }
```

4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b256e78414cf66d97c2907'),
    '1': ObjectId('69b256e78414cf66d97c2908'),
    '2': ObjectId('69b256e78414cf66d97c2909'),
    '3': ObjectId('69b256e78414cf66d97c290a'),
    '4': ObjectId('69b256e78414cf66d97c290b')
  }
}
```

5. Add 5 four-wheeler details to the collection named ‘four_wheelers’. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b257428414cf66d97c290c'),
    '1': ObjectId('69b257428414cf66d97c290d'),
    '2': ObjectId('69b257428414cf66d97c290e'),
    '3': ObjectId('69b257428414cf66d97c290f'),
    '4': ObjectId('69b257428414cf66d97c2910')
  }
}
```

6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.

```
vehicles> db.two_wheelers.find()
[
  {
    _id: ObjectId('69b17dce8f096f66c67c2907'),
    bike_name: 'Honda Shine',
    model: 'gear',
    category: '125cc',
    colors_available: [ 'red', 'black', 'blue' ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: '2020',
    price: 80000
  },
  {
    _id: ObjectId('69b17dce8f096f66c67c2908'),
    bike_name: 'TVS Jupiter',
    model: 'gearless',
    category: '110cc',
    colors_available: [ 'white', 'black' ],
    manufacturer: 'TVS',
    performance: 7,
    timestamp: '2019',
    price: 75000
  },
  {
    _id: ObjectId('69b17dce8f096f66c67c2909'),
    bike_name: 'Yamaha FZ',
    model: 'gear',
    category: '150cc',
    colors_available: [ 'blue', 'black' ],
    manufacturer: 'Yamaha',
    performance: 9,
    timestamp: '2021',
```

```
    performance: 9,
    timestamp: '2021',
    price: 110000
  },
  {
    _id: ObjectId('69b17dce8f096f66c67c290a'),
    bike_name: 'Bajaj Pulsar',
    model: 'gear',
    category: '150cc',
    colors_available: [ 'red', 'black' ],
    manufacturer: 'Bajaj',
    performance: 8,
    timestamp: '2022',
    price: 120000
  },
  {
    _id: ObjectId('69b17dce8f096f66c67c290b'),
    bike_name: 'Suzuki Access',
    model: 'gearless',
    category: '125cc',
    colors_available: [ 'silver', 'white' ],
    manufacturer: 'Suzuki',
    performance: 7,
    timestamp: '2020',
    price: 85000
  },
  {
    _id: ObjectId('69b17e89dcd93221f37c2907'),
    bike_name: 'Honda Shine',
    model: 'gear',
    category: '125cc',
    colors_available: [ 'red', 'black', 'blue' ],
    manufacturer: 'Honda',
```

```
  colors_available: [ 'red', 'black', 'blue' ],
  manufacturer: 'Honda',
  performance: 8,
  timestamp: ISODate('2022-05-10T00:00:00.000Z'),
  price: 85000
},
{
  _id: ObjectId('69b17e89dcd93221f37c2908'),
  bike_name: 'TVS Apache',
  model: 'gear',
  category: '150cc',
  colors_available: [ 'black', 'sport red' ],
  manufacturer: 'TVS',
  performance: 9,
  timestamp: ISODate('2023-01-12T00:00:00.000Z'),
  price: 110000
},
{
  _id: ObjectId('69b17e89dcd93221f37c2909'),
  bike_name: 'Activa 6G',
  model: 'gearless',
  category: '100cc',
  colors_available: [ 'red', 'blue', 'white' ],
  manufacturer: 'Honda',
  performance: 8,
  timestamp: ISODate('2021-08-15T00:00:00.000Z'),
  price: 78000
},
{
```

```
{
  _id: ObjectId('69b17e89dcd93221f37c290a'),
  bike_name: 'Yamaha FZ',
  model: 'gear',
  category: '150cc',
  colors_available: [ 'black', 'blue' ],
  manufacturer: 'Yamaha',
  performance: 9,
  timestamp: ISODate('2022-03-20T00:00:00.000Z'),
  price: 120000
},
{
  _id: ObjectId('69b17e89dcd93221f37c290b'),
  bike_name: 'Suzuki Access',
  model: 'gearless',
  category: '125cc',
  colors_available: [ 'white', 'grey' ],
  manufacturer: 'Suzuki',
  performance: 7,
  timestamp: ISODate('2020-11-11T00:00:00.000Z'),
  price: 82000
},
{
  _id: ObjectId('69b18045f7b1255e457c2907'),
  bike_name: 'Honda Shine',
  model: 'gear',
  category: '125cc',
  colors_available: [ 'red', 'black', 'blue' ],
  manufacturer: 'Honda',
  performance: 8,
  timestamp: '2020',
  price: 80000
},
```

```
},
{
  _id: ObjectId('69b18045f7b1255e457c2908'),
  bike_name: 'TVS Jupiter',
  model: 'gearless',
  category: '110cc',
  colors_available: [ 'white', 'black' ],
  manufacturer: 'TVS',
  performance: 7,
  timestamp: '2019',
  price: 75000
},
{
  _id: ObjectId('69b18045f7b1255e457c2909'),
  bike_name: 'Yamaha FZ',
  model: 'gear',
  category: '150cc',
  colors_available: [ 'blue', 'black' ],
  manufacturer: 'Yamaha',
  performance: 9,
  timestamp: '2021',
  price: 110000
},
{
  _id: ObjectId('69b18045f7b1255e457c290a'),
  bike_name: 'Bajaj Pulsar',
  model: 'gear',
  category: '150cc',
  colors_available: [ 'red', 'black' ],
  manufacturer: 'Bajaj',
  performance: 8,
  timestamp: '2022',
  price: 120000
```

```
},
{
  _id: ObjectId('69b18045f7b1255e457c290b'),
  bike_name: 'Suzuki Access',
  model: 'gearless',
  category: '125cc',
  colors_available: [ 'silver', 'white' ],
  manufacturer: 'Suzuki',
  performance: 7,
  timestamp: '2020',
  price: 85000
}
```

```
vehicles> db.four_wheelers.find()
[
  {
    _id: ObjectId('69b17dda8f096f66c67c290c'),
    vehicle_name: 'Swift',
    model: 'own',
    category: 'car',
    variants: [ 'vxi', 'zxi', 'petrol' ],
    manufacturer: 'Maruti',
    performance: 8,
    timestamp: '2021',
    price: 700000
  },
  {
    _id: ObjectId('69b17dda8f096f66c67c290d'),
    vehicle_name: 'Innova',
    model: 'commercial',
    category: 'car',
    variants: [ 'diesel', 'zxi' ],
    manufacturer: 'Toyota',
    performance: 9,
    timestamp: '2020',
    price: 1500000
  },
  {
    _id: ObjectId('69b17dda8f096f66c67c290e'),
    vehicle_name: 'Tata Bus',
    model: 'commercial',
    category: 'bus',
    variants: [ 'diesel' ],
    manufacturer: 'Tata',
    performance: 7,
    timestamp: '2019',
```

```
    performance: 7,
    timestamp: '2019',
    price: 2000000
  },
  {
    _id: ObjectId('69b17dda8f096f66c67c290f'),
    vehicle_name: 'Mahindra Truck',
    model: 'commercial',
    category: 'heavy truck',
    variants: [ 'diesel' ],
    manufacturer: 'Mahindra',
    performance: 8,
    timestamp: '2018',
    price: 2500000
  },
  {
    _id: ObjectId('69b17dda8f096f66c67c2910'),
    vehicle_name: 'Eicher Lorry',
    model: 'commercial',
    category: 'lorry',
    variants: [ 'diesel' ],
    manufacturer: 'Eicher',
    performance: 7,
    timestamp: '2019',
    price: 1800000
  },
  {
    _id: ObjectId('69b17e92dcd93221f37c290c'),
    vehicle_name: 'Maruti Swift',
    model: 'own',
    category: 'car',
    variants: [ 'vxi', 'zxi', 'petrol' ],
    manufacturer: 'Maruti',
```

```
    variants: [ 'vxi', 'zxi', 'petrol' ],
    manufacturer: 'Maruti',
    performance: 8,
    timestamp: ISODate('2022-06-10T00:00:00.000Z'),
    price: 700000
  },
  {
    _id: ObjectId('69b17e92dcd93221f37c290d'),
    vehicle_name: 'Tata Bus',
    model: 'commercial',
    category: 'bus',
    variants: [ 'diesel' ],
    manufacturer: 'Tata',
    performance: 7,
    timestamp: ISODate('2021-02-15T00:00:00.000Z'),
    price: 2500000
  },
  {
    _id: ObjectId('69b17e92dcd93221f37c290e'),
    vehicle_name: 'Mahindra Bolero',
    model: 'own',
    category: 'car',
    variants: [ 'diesel', 'zxi' ],
    manufacturer: 'Mahindra',
    performance: 8,
    timestamp: ISODate('2023-01-05T00:00:00.000Z'),
    price: 950000
  },
  {
    _id: ObjectId('69b17e92dcd93221f37c290f'),
    vehicle_name: 'Ashok Leyland Truck',
    model: 'commercial',
    category: 'heavy truck',
```

```
    _id: ObjectId('69b17e92dcd93221f37c290f'),
    vehicle_name: 'Ashok Leyland Truck',
    model: 'commercial',
    category: 'heavy truck',
    variants: [ 'diesel' ],
    manufacturer: 'Ashok Leyland',
    performance: 7,
    timestamp: ISODate('2020-09-18T00:00:00.000Z'),
    price: 3200000
  },
  {
    _id: ObjectId('69b17e92dcd93221f37c2910'),
    vehicle_name: 'Eicher Mini Truck',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'diesel' ],
    manufacturer: 'Eicher',
    performance: 6,
    timestamp: ISODate('2022-04-22T00:00:00.000Z'),
    price: 1200000
  },
  {
    _id: ObjectId('69b18181f7b1255e457c290c'),
    vehicle_name: 'Swift',
    model: 'own',
    category: 'car',
    variants: [ 'vxi', 'zxi', 'petrol' ],
    manufacturer: 'Maruti',
    performance: 8,
    timestamp: '2021',
    price: 700000
  },
  {
```

```
},
{
  _id: ObjectId('69b18181f7b1255e457c290d'),
  vehicle_name: 'Innova',
  model: 'commercial',
  category: 'car',
  variants: [ 'diesel', 'zxi' ],
  manufacturer: 'Toyota',
  performance: 9,
  timestamp: '2020',
  price: 1500000
},
{
  _id: ObjectId('69b18181f7b1255e457c290e'),
  vehicle_name: 'Tata Bus',
  model: 'commercial',
  category: 'bus',
  variants: [ 'diesel' ],
  manufacturer: 'Tata',
  performance: 7,
  timestamp: '2019',
  price: 2000000
},
{
  _id: ObjectId('69b18181f7b1255e457c290f'),
  vehicle_name: 'Mahindra Truck',
  model: 'commercial',
  category: 'heavy truck',
  variants: [ 'diesel' ],
  manufacturer: 'Mahindra',
  performance: 8,
  timestamp: '2018',
  price: 2500000
```

```
  },
  {
    _id: ObjectId('69b18181f7b1255e457c2910'),
    vehicle_name: 'Eicher Lorry',
    model: 'commercial',
    category: 'lorry',
    variants: [ 'diesel' ],
    manufacturer: 'Eicher',
    performance: 7,
    timestamp: '2019',
    price: 1800000
  }
]
```

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database

```
vehicles> db.two_wheelers.find({}, {bike_name:1,price:1,_id:0})
[
  { bike_name: 'Honda Shine', price: 80000 },
  { bike_name: 'TVS Jupiter', price: 75000 },
  { bike_name: 'Yamaha FZ', price: 110000 },
  { bike_name: 'Bajaj Pulsar', price: 120000 },
  { bike_name: 'Suzuki Access', price: 85000 },
  { bike_name: 'Honda Shine', price: 85000 },
  { bike_name: 'TVS Apache', price: 110000 },
  { bike_name: 'Activa 6G', price: 78000 },
  { bike_name: 'Yamaha FZ', price: 120000 },
  { bike_name: 'Suzuki Access', price: 82000 },
  { bike_name: 'Honda Shine', price: 80000 },
  { bike_name: 'TVS Jupiter', price: 75000 },
  { bike_name: 'Yamaha FZ', price: 110000 },
  { bike_name: 'Bajaj Pulsar', price: 120000 },
  { bike_name: 'Suzuki Access', price: 85000 }
]
```

```
vehicles> db.four_wheelers.find({}, {vehicle_name:1,price:1,_id:0})
[
  { vehicle_name: 'Swift', price: 700000 },
  { vehicle_name: 'Innova', price: 1500000 },
  { vehicle_name: 'Tata Bus', price: 2000000 },
  { vehicle_name: 'Mahindra Truck', price: 2500000 },
  { vehicle_name: 'Eicher Lorry', price: 1800000 },
  { vehicle_name: 'Maruti Swift', price: 700000 },
  { vehicle_name: 'Tata Bus', price: 2500000 },
  { vehicle_name: 'Mahindra Bolero', price: 950000 },
  { vehicle_name: 'Ashok Leyland Truck', price: 3200000 },
  { vehicle_name: 'Eicher Mini Truck', price: 1200000 },
  { vehicle_name: 'Swift', price: 700000 },
  { vehicle_name: 'Innova', price: 1500000 },
  { vehicle_name: 'Tata Bus', price: 2000000 },
  { vehicle_name: 'Mahindra Truck', price: 2500000 },
  { vehicle_name: 'Eicher Lorry', price: 1800000 }
]
```

8. Write a MongoDB query to display two_wheelers from a particular company

```
vehicles> db.two_wheelers.find({manufacturer:"Honda"})
[
  {
    _id: ObjectId('69b17dce8f096f66c67c2907'),
    bike_name: 'Honda Shine',
    model: 'gear',
    category: '125cc',
    colors_available: [ 'red', 'black', 'blue' ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: '2020',
    price: 80000
  },
  {
    _id: ObjectId('69b17e89dcd93221f37c2907'),
    bike_name: 'Honda Shine',
    model: 'gear',
    category: '125cc',
    colors_available: [ 'red', 'black', 'blue' ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: ISODate('2022-05-10T00:00:00.000Z'),
    price: 85000
  },
  {
    _id: ObjectId('69b17e89dcd93221f37c2909'),
    bike_name: 'Activa 6G',
    model: 'gearless',
    category: '100cc',
    colors_available: [ 'red', 'blue', 'white' ],
    manufacturer: 'Honda',
    performance: 8,
```

```
    performance: 8,
    timestamp: ISODate('2021-08-15T00:00:00.000Z'),
    price: 78000
  },
  {
    _id: ObjectId('69b18045f7b1255e457c2907'),
    bike_name: 'Honda Shine',
    model: 'gear',
    category: '125cc',
    colors_available: [ 'red', 'black', 'blue' ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: '2020',
    price: 80000
  }
]
```

9.Write a MongoDB query to display four_wheelers available in diesel variants

```
vehicles> db.four_wheelers.find({variants:"diesel"})
[
  {
    _id: ObjectId('69b17dda8f096f66c67c290d'),
    vehicle_name: 'Innova',
    model: 'commercial',
    category: 'car',
    variants: [ 'diesel', 'zxi' ],
    manufacturer: 'Toyota',
    performance: 9,
    timestamp: '2020',
    price: 1500000
  },
  {
    _id: ObjectId('69b17dda8f096f66c67c290e'),
    vehicle_name: 'Tata Bus',
    model: 'commercial',
    category: 'bus',
    variants: [ 'diesel' ],
    manufacturer: 'Tata',
    performance: 7,
    timestamp: '2019',
    price: 2000000
  },
  {
    _id: ObjectId('69b17dda8f096f66c67c290f'),
    vehicle_name: 'Mahindra Truck',
    model: 'commercial',
    category: 'heavy truck',
    variants: [ 'diesel' ],
    manufacturer: 'Mahindra',
    performance: 8,
    timestamp: '2018',
```

```
    timestamp: '2018',
    price: 2500000
  },
  {
    _id: ObjectId('69b17dda8f096f66c67c2910'),
    vehicle_name: 'Eicher Lorry',
    model: 'commercial',
    category: 'lorry',
    variants: [ 'diesel' ],
    manufacturer: 'Eicher',
    performance: 7,
    timestamp: '2019',
    price: 1800000
  },
  {
    _id: ObjectId('69b17e92dcd93221f37c290d'),
    vehicle_name: 'Tata Bus',
    model: 'commercial',
    category: 'bus',
    variants: [ 'diesel' ],
    manufacturer: 'Tata',
    performance: 7,
    timestamp: ISODate('2021-02-15T00:00:00.000Z'),
    price: 2500000
  },
  {
    _id: ObjectId('69b17e92dcd93221f37c290e'),
    vehicle_name: 'Mahindra Bolero',
    model: 'own',
    category: 'car',
    variants: [ 'diesel', 'zxi' ],
    manufacturer: 'Mahindra',
    performance: 8,
```

```
    manufacturer: 'Mahindra',
    performance: 8,
    timestamp: ISODate('2023-01-05T00:00:00.000Z'),
    price: 950000
  },
  {
    _id: ObjectId('69b17e92dcd93221f37c290f'),
    vehicle_name: 'Ashok Leyland Truck',
    model: 'commercial',
    category: 'heavy truck',
    variants: [ 'diesel' ],
    manufacturer: 'Ashok Leyland',
    performance: 7,
    timestamp: ISODate('2020-09-18T00:00:00.000Z'),
    price: 3200000
  },
  {
    _id: ObjectId('69b17e92dcd93221f37c2910'),
    vehicle_name: 'Eicher Mini Truck',
    model: 'commercial',
    category: 'mini truck',
    variants: [ 'diesel' ],
    manufacturer: 'Eicher',
    performance: 6,
    timestamp: ISODate('2022-04-22T00:00:00.000Z'),
    price: 1200000
  },
  {
    _id: ObjectId('69b18181f7b1255e457c290d'),
    vehicle_name: 'Innova',
    model: 'commercial',
    category: 'car',
    variants: [ 'diesel', 'zxi' ],
```

```
    category: 'car',
    variants: [ 'diesel', 'zxi' ],
    manufacturer: 'Toyota',
    performance: 9,
    timestamp: '2020',
    price: 1500000
  },
  {
    _id: ObjectId('69b18181f7b1255e457c290e'),
    vehicle_name: 'Tata Bus',
    model: 'commercial',
    category: 'bus',
    variants: [ 'diesel' ],
    manufacturer: 'Tata',
    performance: 7,
    timestamp: '2019',
    price: 2000000
  },
  {
    _id: ObjectId('69b18181f7b1255e457c290f'),
    vehicle_name: 'Mahindra Truck',
    model: 'commercial',
    category: 'heavy truck',
    variants: [ 'diesel' ],
    manufacturer: 'Mahindra',
    performance: 8,
    timestamp: '2018',
    price: 2500000
  },
  {
    _id: ObjectId('69b18181f7b1255e457c2910'),
    vehicle_name: 'Eicher Lorry',
    model: 'commercial',
```

```
  {
    _id: ObjectId('69b18181f7b1255e457c2910'),
    vehicle_name: 'Eicher Lorry',
    model: 'commercial',
    category: 'lorry',
    variants: [ 'diesel' ],
    manufacturer: 'Eicher',
    performance: 7,
    timestamp: '2019',
    price: 1800000
  }
]
```

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

```
vehicles> db.two_wheelers.find(
| {performance:{$gt:5}},
| {bike_name:1,category:1,manufacturer:1,_id:0}
| )
[
  { bike_name: 'Honda Shine', category: '125cc', manufacturer: 'Honda' },
  { bike_name: 'TVS Jupiter', category: '110cc', manufacturer: 'TVS' },
  { bike_name: 'Yamaha FZ', category: '150cc', manufacturer: 'Yamaha' },
  { bike_name: 'Bajaj Pulsar', category: '150cc', manufacturer: 'Bajaj' },
  {
    bike_name: 'Suzuki Access',
    category: '125cc',
    manufacturer: 'Suzuki'
  },
  { bike_name: 'Honda Shine', category: '125cc', manufacturer: 'Honda' },
  { bike_name: 'TVS Apache', category: '150cc', manufacturer: 'TVS' },
  { bike_name: 'Activa 6G', category: '100cc', manufacturer: 'Honda' },
  { bike_name: 'Yamaha FZ', category: '150cc', manufacturer: 'Yamaha' },
  {
    bike_name: 'Suzuki Access',
    category: '125cc',
    manufacturer: 'Suzuki'
  },
  { bike_name: 'Honda Shine', category: '125cc', manufacturer: 'Honda' },
  { bike_name: 'TVS Jupiter', category: '110cc', manufacturer: 'TVS' },
  { bike_name: 'Yamaha FZ', category: '150cc', manufacturer: 'Yamaha' },
  { bike_name: 'Bajaj Pulsar', category: '150cc', manufacturer: 'Bajaj' },
  {
    bike_name: 'Suzuki Access',
    category: '125cc',
    manufacturer: 'Suzuki'
  }
]
```

```
vehicles> db.four_wheelers.find(
| {performance:{$gt:5}},
| {vehicle_name:1,category:1,manufacturer:1,_id:0}
| )
[
  { vehicle_name: 'Swift', category: 'car', manufacturer: 'Maruti' },
  { vehicle_name: 'Innova', category: 'car', manufacturer: 'Toyota' },
  { vehicle_name: 'Tata Bus', category: 'bus', manufacturer: 'Tata' },
  {
    vehicle_name: 'Mahindra Truck',
    category: 'heavy truck',
    manufacturer: 'Mahindra'
  },
  {
    vehicle_name: 'Eicher Lorry',
    category: 'lorry',
    manufacturer: 'Eicher'
  },
  {
    vehicle_name: 'Maruti Swift',
    category: 'car',
    manufacturer: 'Maruti'
  },
  { vehicle_name: 'Tata Bus', category: 'bus', manufacturer: 'Tata' },
  {
    vehicle_name: 'Mahindra Bolero',
    category: 'car',
    manufacturer: 'Mahindra'
  },
  {
    vehicle_name: 'Ashok Leyland Truck',
    category: 'heavy truck',
    manufacturer: 'Ashok Leyland'
```

```
  {
    vehicle_name: 'Ashok Leyland Truck',
    category: 'heavy truck',
    manufacturer: 'Ashok Leyland'
  },
  {
    vehicle_name: 'Eicher Mini Truck',
    category: 'mini truck',
    manufacturer: 'Eicher'
  },
  { vehicle_name: 'Swift', category: 'car', manufacturer: 'Maruti' },
  { vehicle_name: 'Innova', category: 'car', manufacturer: 'Toyota' },
  { vehicle_name: 'Tata Bus', category: 'bus', manufacturer: 'Tata' },
  {
    vehicle_name: 'Mahindra Truck',
    category: 'heavy truck',
    manufacturer: 'Mahindra'
  },
  {
    vehicle_name: 'Eicher Lorry',
    category: 'lorry',
    manufacturer: 'Eicher'
  }
]
```

2. Use MongoDB to implement the following DB operations for a Zoo

1. Create a database called ‘animal’ and write a MongoDB query to select database as ‘animal’.

```
test> use animal
switched to db animal
```

2. Write a MongoDB query to display all the databases.

```
animal> show dbs
admin       40.00 KiB
config     100.00 KiB
local       64.00 KiB
vehicles   144.00 KiB
```

3. Create a collection called ‘wild_animals’.(use capping) and Create a collection called ‘domestic_animals’.

```
animal> db.createCollection("wild_animals",{capped:true,size:50000})
{ ok: 1 }
animal> db.createCollection("domestic_animals")
{ ok: 1 }
```

4. Add 5 wild_animal details to the collection named ‘wild_animals’. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b26b5802fd2fa8b87c2907'),
    '1': ObjectId('69b26b5802fd2fa8b87c2908'),
    '2': ObjectId('69b26b5802fd2fa8b87c2909'),
    '3': ObjectId('69b26b5802fd2fa8b87c290a'),
    '4': ObjectId('69b26b5802fd2fa8b87c290b')
  }
}
```

5. Add 5 domestic-animal details to the collection named ‘domestic_animals’. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b26ba802fd2fa8b87c290c'),
    '1': ObjectId('69b26ba802fd2fa8b87c290d'),
    '2': ObjectId('69b26ba802fd2fa8b87c290e'),
    '3': ObjectId('69b26ba802fd2fa8b87c290f'),
    '4': ObjectId('69b26ba802fd2fa8b87c2910')
  }
}
```

6. Write a MongoDB query to display all documents available in wild_animals and domastic_animals.

```
animal> db.wild_animals.find()
[
  {
    _id: ObjectId('69b26b5802fd2fa8b87c2907'),
    animal_name: 'Lion',
    nature: 'harm',
    favorite_foods: [ 'meat', 'deer' ],
    care_taker_name: 'Ramesh',
    life_span: 15,
    timestamp: '2022',
    expenses: 50000
  },
  {
    _id: ObjectId('69b26b5802fd2fa8b87c2908'),
    animal_name: 'Tiger',
    nature: 'harm',
    favorite_foods: [ 'meat', 'rabbits' ],
    care_taker_name: 'Suresh',
    life_span: 18,
    timestamp: '2021',
    expenses: 60000
  },
  {
    _id: ObjectId('69b26b5802fd2fa8b87c2909'),
    animal_name: 'Elephant',
    nature: 'harmless',
    favorite_foods: [ 'grass', 'fruits' ],
    care_taker_name: 'Mahesh',
    life_span: 60,
    timestamp: '2020',
    expenses: 70000
  },
  {
```

```
  {
    _id: ObjectId('69b26b5802fd2fa8b87c290a'),
    animal_name: 'Deer',
    nature: 'harmless',
    favorite_foods: [ 'grass', 'plants' ],
    care_taker_name: 'Ramesh',
    life_span: 12,
    timestamp: '2019',
    expenses: 20000
  },
  {
    _id: ObjectId('69b26b5802fd2fa8b87c290b'),
    animal_name: 'Bear',
    nature: 'harm',
    favorite_foods: [ 'fish', 'honey' ],
    care_taker_name: 'Kiran',
    life_span: 20,
    timestamp: '2023',
    expenses: 45000
  }
]
```

```
animal> db.domestic_animals.find()
[
  {
    _id: ObjectId('69b26ba802fd2fa8b87c290c'),
    animal_name: 'Dog',
    gender: 'male',
    favorite_foods: [ 'meat', 'rice' ],
    animal_petname: 'Tommy',
    life_span: 12,
    timestamp: '2022',
    expenses: 10000
  },
  {
    _id: ObjectId('69b26ba802fd2fa8b87c290d'),
    animal_name: 'Cat',
    gender: 'female',
    favorite_foods: [ 'fish', 'milk' ],
    animal_petname: 'Kitty',
    life_span: 15,
    timestamp: '2021',
    expenses: 8000
  },
  {
    _id: ObjectId('69b26ba802fd2fa8b87c290e'),
    animal_name: 'Cow',
    gender: 'female',
    favorite_foods: [ 'grass', 'fodder' ],
    animal_petname: 'Lakshmi',
    life_span: 20,
    timestamp: '2020',
    expenses: 12000
  },
```

```
  {
    _id: ObjectId('69b26ba802fd2fa8b87c290f'),
    animal_name: 'Goat',
    gender: 'male',
    favorite_foods: [ 'leaves', 'grass' ],
    animal_petname: 'Bunny',
    life_span: 10,
    timestamp: '2019',
    expenses: 7000
  },
  {
    _id: ObjectId('69b26ba802fd2fa8b87c2910'),
    animal_name: 'Horse',
    gender: 'male',
    favorite_foods: [ 'grass', 'grains' ],
    animal_petname: 'Rocky',
    life_span: 25,
    timestamp: '2023',
    expenses: 15000
  }
]
```

7. Write a MongoDB query to display only animal name and expenses in all the collection of the database.

```
animal> db.wild_animals.find({}, {animal_name:1, expenses:1, _id:0})
[
  { animal_name: 'Lion', expenses: 50000 },
  { animal_name: 'Tiger', expenses: 60000 },
  { animal_name: 'Elephant', expenses: 70000 },
  { animal_name: 'Deer', expenses: 20000 },
  { animal_name: 'Bear', expenses: 45000 }
]
```

```
animal> db.domestic_animals.find({}, {animal_name:1, expenses:1, _id:0})
[
  { animal_name: 'Dog', expenses: 10000 },
  { animal_name: 'Cat', expenses: 8000 },
  { animal_name: 'Cow', expenses: 12000 },
  { animal_name: 'Goat', expenses: 7000 },
  { animal_name: 'Horse', expenses: 15000 }
]
```

8. Write a MongoDB query to display domestic_animals whose life is a particular year.

```
animal> db.domestic_animals.find({life_span:12})
[
  {
    _id: ObjectId('69b26ba802fd2fa8b87c290c'),
    animal_name: 'Dog',
    gender: 'male',
    favorite_foods: [ 'meat', 'rice' ],
    animal_petname: 'Tommy',
    life_span: 12,
    timestamp: '2022',
    expenses: 10000
  }
]
```

9. Write a MongoDB query to display wild_animals available under a particular care_taker.

```
animal> db.wild_animals.find({care_taker_name:"Ramesh"})
[
  {
    _id: ObjectId('69b26b5802fd2fa8b87c2907'),
    animal_name: 'Lion',
    nature: 'harm',
    favorite_foods: [ 'meat', 'deer' ],
    care_taker_name: 'Ramesh',
    life_span: 15,
    timestamp: '2022',
    expenses: 50000
  },
  {
    _id: ObjectId('69b26b5802fd2fa8b87c290a'),
    animal_name: 'Deer',
    nature: 'harmless',
    favorite_foods: [ 'grass', 'plants' ],
    care_taker_name: 'Ramesh',
    life_span: 12,
    timestamp: '2019',
    expenses: 20000
  }
]
```

10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

```
animal> db.wild_animals.find(
| {life_span:{$gt:5}},
| {animal_name:1,favorite_foods:1,expenses:1,_id:0}
| )
[
  {
    animal_name: 'Lion',
    favorite_foods: [ 'meat', 'deer' ],
    expenses: 50000
  },
  {
    animal_name: 'Tiger',
    favorite_foods: [ 'meat', 'rabbits' ],
    expenses: 60000
  },
  {
    animal_name: 'Elephant',
    favorite_foods: [ 'grass', 'fruits' ],
    expenses: 70000
  },
  {
    animal_name: 'Deer',
    favorite_foods: [ 'grass', 'plants' ],
    expenses: 20000
  },
  {
    animal_name: 'Bear',
    favorite_foods: [ 'fish', 'honey' ],
    expenses: 45000
  }
]
```

```
animal> db.domestic_animals.find(
| {life_span:{$gt:5}},
| {animal_name:1,favorite_foods:1,expenses:1,_id:0}
| )
[
  {
    animal_name: 'Dog',
    favorite_foods: [ 'meat', 'rice' ],
    expenses: 10000
  },
  {
    animal_name: 'Cat',
    favorite_foods: [ 'fish', 'milk' ],
    expenses: 8000
  },
  {
    animal_name: 'Cow',
    favorite_foods: [ 'grass', 'fodder' ],
    expenses: 12000
  },
  {
    animal_name: 'Goat',
    favorite_foods: [ 'leaves', 'grass' ],
    expenses: 7000
  },
  {
    animal_name: 'Horse',
    favorite_foods: [ 'grass', 'grains' ],
    expenses: 15000
  }
]
```